LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

รายชื่อสมาชิก

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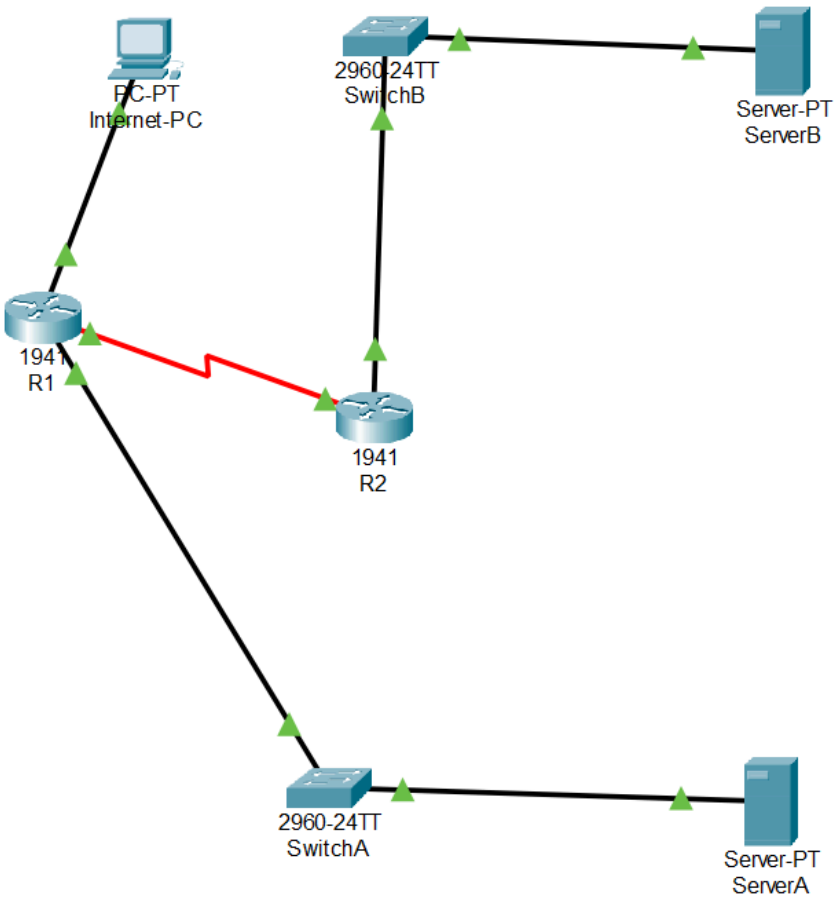
TOPOLOGY

None

```
                INTERNET
                    |
               (DHCP)
                R1 G0/0
                    |
            [NAT OUTSIDE]
                    |
           R1 S0/0/0
        100.10.10.1/30
                    |
           R2 S0/0/0
        100.10.10.2/30
                    |
-----------------------
```

```
          |                        |
R1 G0/1 (LAN A)        R2 G0/1 (LAN B)
192.168.10.1            192.168.20.1
          |                        |
     ServerA                 ServerB
Week03 Microservices   Week03 Microservices
```

ผลลัพธ์ :



ADDRESS TABLE

Device	Interface	IP	Role
R1	G0/0	DHCP	Internet
R1	S0/0/0	100.10.10.1/30	ISP
R1	G0/1	192.168.10.1/24	LAN A
R2	S0/0/0	100.10.10.2/30	ISP
R2	G0/1	192.168.20.1/24	LAN B
ServerA	eth0	192.168.10.10	Microservices A
ServerB	eth0	192.168.20.10	Microservices B

PART 1 CISCO CONFIGURATION (STRICT)

R1 — Internet Edge Router

```
None
enable
conf t
hostname R1

interface g0/0
 ip address dhcp
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
```

```
 no shutdown

interface s0/0/0
 ip address 100.10.10.1 255.255.255.252
 clock rate 64000
 no shutdown

ip route 192.168.20.0 255.255.255.0 100.10.10.2
ip route 0.0.0.0 0.0.0.0 g0/0

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

---

step 1 : เพิ่ม port s0/0/0 ให้กับ R1 เพื่อให้ตรงกับ Configของ Lab โดยใช้ HWIC-2T ใส่ไว้ที่ Port 0 ของ R

Step 2 : จากนั้นทำการ Config R1

```
R1>enable
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#interface g0/0
R1(config-if)#ip address dhcp
R1(config-if)#ip nat outside
R1(config-if)#on shutdown
             ^
% Invalid input detected at '^' marker.

R1(config-if)#no shutdown
R1(config-if)#interface g0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#ip nat inside
R1(config-if)#no shutdown
R1(config-if)#
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

```
R1(config-if)#interface s0/0/0
R1(config-if)#ip address 100.10.10.1 255.255.255.252
R1(config-if)#clock rate 64000
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#write memory
Building configuration...
[OK]
```

R2 — Branch Router

```
None
enable
conf t
hostname R2

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface s0/0/0
 ip address 100.10.10.2 255.255.255.252
 no shutdown

ip route 192.168.10.0 255.255.255.0 100.10.10.1
ip route 0.0.0.0 0.0.0.0 100.10.10.1

end
write memory
```

step 1 : เช่นเดียวกับ R1 ต้องเพิ่ม port s0/0/0 ให้กับ R2 เพื่อให้ตรงกับ Config Code ของ Lab

Step 2 : จากนั้นทำการ Config R2

```
R2>enable
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#interface g0/1
R2(config-if)#ip address 192.168.20 255.255.255.0

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#interface s0/0/0
R2(config-if)#ip address 100.10.10.2 255.255.255.252
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R2(config-if)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#write memory
Building configuration...
[OK]
```

PART 2 — Deploy Week03 Microservices

Based on your Week03 architecture from `PROJECT_PROFILE.md`

We deploy:

- Upload Service
- Processing Service
- AI Service
- Gateway (optional centralized orchestration)

🛠️ On ServerA (LAN A)

```shell
cd automation/week03-microservices/phase1
python start_services.py
```

ผลลัพธ์ :

```
PS E:\network\automation\week03-microservices\phase1> py start_services.py
🎯 Starting Phase 1 Microservices...
==============================================
🚀 Starting Upload Service on port 8001...
⏳ Waiting for Upload Service to become ready...
✅ Upload Service is ready!
🚀 Starting Processing Service on port 8002...
⏳ Waiting for Processing Service to become ready...
✅ Processing Service is ready!
🚀 Starting AI Service on port 8003...
⏳ Waiting for AI Service to become ready...
✅ AI Service is ready!
🚀 Starting Gateway Service on port 8090...
⏳ Waiting for Gateway Service to become ready...
✅ Gateway Service is ready!

🎉 All services are running!

🔍 Checking system health...
System status: healthy
  ✅ gateway: healthy
  ✅ upload: healthy
  ✅ processing: healthy
  ✅ ai: healthy

🧪 Running quick integration test...
📁 Created test file: test_data\sample.jpg
❌ Quick test failed: 422
   Response: {"detail":[{"type":"model_attributes_type","loc":["body","processing_options"],"msg":"Input should be a valid dictionary or object
 to extract fields from","input":"{\"enable_processing\": true, \"processing_operation\": \"thumbnail\", \"enable_ai_analysis\": true, \"ai_ana
lysis_type\": \"vision\"}"}]}

🎯 Phase 1 Microservices are ready!

Available endpoints:
  Gateway:    http://localhost:8090
  Upload:     http://localhost:8001
  Processing: http://localhost:8002
  AI:         http://localhost:8003
```

```
Test commands:
  curl -X POST -F 'file=@test_data/sample.jpg' http://localhost:8090/process-file
  pytest tests/test_phase1_compliance.py -v

🔄 Press Ctrl+C to stop all services...

📊 Service Status:
----------------------------------------
Upload Service       🟢 Running
Processing Service   🟢 Running
AI Service           🟢 Running
Gateway Service      🟢 Running
```

Services:

- Gateway→8090
- Upload→8001
- Processing→8002
- AI→8003

ServerA IP:

```
None
192.168.10.10
```

🛠️ On ServerB (LAN B)

Same deployment:

```
Shell
python start_services.py
```

เนื่องจากเราใช้การจำลองใน simulate ที่เครื่องเดียวกัน ทำให้ไม่สามารถทำขั้นตอนนี้ได้

```
None
192.168.20.10
```

🧪 PART 3 — WAN + INTERNET TESTING

🔵 Test 1 — Cross-Site Upload (LAN A → LAN B)

From ClientA:

```
None
curl http://192.168.20.10:8000/health
```

Traffic should traverse:

ClientA (LAN A) → R1 (Default Gateway) → Serial WAN (100.10.10.0/30) → R2 → ServerB (LAN B)

```
PS E:\network\automation\week03-microservices\phase1> curl http://localhost:8080/health

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
      RECOMMENDED ACTION:
      Use the -UseBasicParsing switch to avoid script code execution.

      Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): Y


StatusCode        : 200
StatusDescription : OK
Content           : {"status":"healthy","timestamp":"2026-03-30T18:08:19.306791","check_duration":0.036809,"services":{"gateway":{"url":"http:
                    //localhost:8090/health","status":"healthy"},"upload":{"url":"http://upload:80...
RawContent        : HTTP/1.1 200 OK
                    Content-Length: 360
                    Content-Type: application/json
                    Date: Mon, 30 Mar 2026 18:08:19 GMT
                    Server: uvicorn

                    {"status":"healthy","timestamp":"2026-03-30T18:08:19.306791","check_durati...
Forms             : {}
Headers           : {[Content-Length, 360], [Content-Type, application/json], [Date, Mon, 30 Mar 2026 18:08:19 GMT], [Server, uvicorn]}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 360
```

🔵 Test 2 — Full Workflow Across ISP

From LAN A client:

```
None
curl -X POST http://192.168.20.10:9000/process-file
```

Expected Flow: Client → Gateway → Upload → Processing → AI → Response

Result :

```
PS E:\network\automation\week03-microservices\phase1> curl.exe -X POST -F "file=@test_data/sample.jpg" http://localhost:8080/process-file
{"workflow_id":"workflow_20260330_184835","file_id":"796251ea-1269-48df-86cf-97b7f735c41c","upload_status":"completed","processing_status":"com
pleted","ai_analysis_status":"success","total_time":6.361278,"timestamp":"2026-03-30T18:48:41.719996"}
```

🔵 Test 3 — Internet Reachability

From ServerA:

```
None
ping 8.8.8.8
```

```
C:\>ping 100.10.10.2

Pinging 100.10.10.2 with 32 bytes of data:

Reply from 100.10.10.2: bytes=32 time=13ms TTL=254
Reply from 100.10.10.2: bytes=32 time=8ms TTL=254
Reply from 100.10.10.2: bytes=32 time=7ms TTL=254
Reply from 100.10.10.2: bytes=32 time=8ms TTL=254

Ping statistics for 100.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 13ms, Average = 9ms
```

Verify NAT overload working on R1:

```
None
show ip nat translations
```

Result :

```
R1>show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 100.10.10.1:17    192.168.10.10:17   100.10.10.2:17     100.10.10.2:17
icmp 100.10.10.1:18    192.168.10.10:18   100.10.10.2:18     100.10.10.2:18
icmp 100.10.10.1:19    192.168.10.10:19   100.10.10.2:19     100.10.10.2:19
icmp 100.10.10.1:20    192.168.10.10:20   100.10.10.2:20     100.10.10.2:20
```

🔎 Verification Commands

On R1

None

```
show ip route
show ip nat translations
show ip interface brief
```

```
R1>show ip nat translations
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/0/0
L       100.10.10.1/32 is directly connected, Serial0/0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 100.10.10.2
```

```
R1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES DHCP   up                    up
GigabitEthernet0/1     192.168.10.1    YES manual up                    up
Serial0/0/0            100.10.10.1     YES manual up                    up
Serial0/0/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

On R2

```
None
show ip route
ping 100.10.10.1
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 100.10.10.1 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/0/0
L       100.10.10.2/32 is directly connected, Serial0/0/0
S    192.168.10.0/24 [1/0] via 100.10.10.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 100.10.10.1
```

```
R2#ping 100.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/9 ms
```

Engineering Reflection (Week03 Integration)

From your Week03 profile :

This lab now demonstrates:

Concept	Where Demonstrated
Public vs Private Boundaries	R1 NAT edge
Distributed Services	ServerA & ServerB
Network Segmentation	Serial ISP /30
Horizontal Architecture	Two independent microservice stacks
Internet Edge Pattern	R1 as NAT Gateway
Service Isolation	LAN A vs LAN B

What Changed from Lab 4

Lab 4	Lab 5
10.10.12.0/30 simulated internet	Real ISP serial 100.10.10.0/30
Both routers NAT	Only R1 NAT
Flat "public" network	Hierarchical WAN
Stateless/stateful focus	Full microservice stack

Architectural Significance

You now have:

- Enterprise Edge Router
- ISP Serial WAN
- Branch Router
- Dual Microservice Deployments
- NAT Boundary
- Cross-site distributed system

This is no longer a classroom NAT lab.

This is a **miniature cloud-edge + branch distributed architecture.**